



CEE (COMMON ENTRANCE EXAMINATION)

Marks distribution Pattern: -

Physics	: -	50
Chemistry	: -	50
Zoology	: -	40
Biology/Botany	: -	40
MAT	: -	20

MBBS , BDS , B.Sc, Nursing
Paramedical / Applied Sciences

Time : 3 hr
Full Mark : 200
Negative marking of 0.25 for each wrong tick mark.

1. The frequency of fork is increased by 50% ,then change in tension is

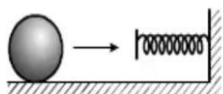
- a)50%. b)100%. c)75% d)125%

2.The sum of 2 vector is also the unit vector ,then their difference will be

- a)2. b) $3^{1/2}$. c) $2^{1/2}$ d)Zero

3.A ball of mass 4 kg moving with speed of 5 m/s on a horizontal smooth surface, collides with a nearly weightless spring of force constant $k=20\text{N/M}$. The maximum compression of spring would be

- a)0.15m. b)1.5m. c)1.8m. d)2.2m



4.A stunt driver wants to jump a car across a 20-meter-wide canyon. He drives up an approach ramp inclined at θ to the horizontal. If his speed at the top of the ramp is 20 m/s, what minimum angle θ ensures he clears the gap safely? (Take $g = 10 \text{ m/s}^2$):

- a) 15° . b) 30° . c) 45° d) 60°

5.A block of mass 2 kg is pressed against a rough vertical wall with a horizontal force of 40 N. If the coefficient of static friction between the block and the wall is 0.5, the frictional force acting on the block is (Take $g = 10 \text{ m/s}^2$):

- a) 20 N b)10 N c) 40 N d) 0 N

6.A force $F = 20 + 10y$ acts on a particle in y-direction where F is in newton and y in meter. Work done by this force to move the particle from $y = 0$ to $y = 1 \text{ m}$ is:

- a) 20 J. b) 30 J. c) 5 J d) 25J

7.A solid cylinder and a solid sphere of equal mass and radius roll down the same inclined plane without slipping from rest. What is the ratio of their linear velocities ($V_{\text{cylinder}}/V_{\text{sphere}}$) at the bottom of the incline?

- a) $\mu(14/15)$. b) $\mu(15/14)$ c) $\mu(10/11)$ d) $\mu(11/10)$

8.If the acceleration due to gravity at the surface of earth is g, then the acceleration due to gravity at an altitude equal to half the radius of the earth ($h = R/2$) is:

- a) $4g/9$. b) $9g/4$. c) $2g/3$ d) $g/2$

9. The excess pressure inside a spherical soap bubble of radius r is balanced by a column of oil of specific gravity 0.8 to a height of 2 mm. If the surface tension of soap solution is 0.04 N/m , the value of r is ($g = 10 \text{ m/s}^2$):

- a) $10 \mu\text{m}$ b) $20 \mu\text{m}$ c) $30 \mu\text{m}$ d) $40 \mu\text{m}$

10.A wire of length L and cross-sectional area A is made of a material of Young's modulus Y. If the wire is stretched by an amount x, the work done in stretching the wire is:

- a) $Y A x^2 / 2L$. b) $Y A x^2 / L$ c) $Y A x / L$ d) $Y A L / 2x^2$

11.A centigrade and a Fahrenheit thermometer are dipped in boiling water. Now water temperature is lowered until the Fahrenheit thermometer registers 140°F . What is the fall in temperature as registered by the centigrade thermometer?

- a) 30°C . b) 60°C . c) 40°C d) 80°C

12. The thermal conductivity of a material depends directly on which of the following physical variables?

- a) Area of cross-section c) Temperature gradient
b) Length of the rod . d) Nature of the material

13. The temperature of an ideal gas is increased from 27°C to 927°C . The root-mean-square (rms) speed of its molecules becomes:

- a) Twice b) Three times c) Four times d) Half

14. An ideal gas undergoes an adiabatic process. If the pressure of the gas is proportional to the cube of its absolute temperature ($P \propto T^3$), the ratio of specific heats ($\gamma = C_p/C_v$) for this gas is:

- a) 1.5. b) 1.33. c) 1.4 d) 1.67

15.A Carnot engine has an efficiency of 50% when its sink is at a temperature of 27°C .

The temperature of the source must be:

- a) 300 K. b) 327 K c) 600 K d) 54°C

16. The value of Joule's mechanical equivalent of heat (J) acts fundamentally as a:

- a) Physical constant. c) Conversion factor
b) Unitless scalar. d) Thermodynamic variable

17. The van der Waals equation of state for 1 mole of a real gas is written as $(P + a/V^2)(V - b) = RT$. The SI units of the critical internal correction constant a are:

- a) $\text{N}\cdot\text{m}^4 / \text{mol}^2$. b) $\text{N}\cdot\text{m} / \text{mol}$. c) $\text{N}\cdot\text{m}^2 / \text{mol}^2$ d) $\text{N}\cdot\text{m}^3 / \text{mol}^2$

18.A string is fixed at both ends and vibrates in its 4th harmonic. If the total length of the string is 2.0 m and the speed of the transverse wave is 100 m/s, the frequency of vibration is:

- a) 25 Hz b) 50 Hz c) 100 Hz d) 200 Hz

19.A source of sound emitting a frequency of 400 Hz moves toward a stationary observer with a speed of 20 m/s. If the speed of sound in air is 340 m/s, the apparent frequency heard by the observer is:

- a) 425 Hz b)377 Hz. c) 400 Hz d) 450 Hz

20.A ray of light is incident at an angle of incidence i on one face of a small angle prism

with a refracting angle A, and emerges normally from the opposite surface. If the refractive index of the prism material is μ , the angle of incidence i is nearly equal to:

- a) A / μ . b) $\mu A / 2$ c) μA d) $A / 2\mu$

21. A biconvex lens has radii of curvature 20 cm and 30 cm respectively. If the refractive index of the material of the lens is 1.5, its focal length is:

- a) 18 cm b) 34 cm c) 24 cm d) 30 cm

22. In a Young's double slit experiment, the slit separation is doubled and the distance between the slits and screen is halved. The fringe width becomes:

- a) One-fourth b) Half c) Four times d) Unchanged

23. An unpolarized beam of light of intensity I_0 is incident on a linear polaroid sheet. The intensity of light transmitted through this initial sheet is:

- a) I_0 . b) Zero c) $I_0 / 2$ d) $I_0 / 4$

24. A person with a near point of 50 cm wants to read a book clearly at a distance of 25 cm. What is the power of the corrective spectacle lens required?

- a) +2.0 D. b) +4.0 D c) -2.0 D d) -4.0 D

25. White light is used to illuminate the two slits in a Young's double slit experiment. The center of the fringe pattern on the screen will appear:

- a) Bright white c) Dark/black
b) Bright red d) Symmetrically blue

26. A carbon resistor has color bands in the following sequence: Yellow, Violet, Brown, and Gold. The resistance value and tolerance percentage are respectively:

- a) $470 \Omega \pm 5\%$. b) $4.7 k\Omega \pm 5\%$. c) $47 \Omega \pm 10\%$ d) $470 \Omega \pm 10\%$

27. A uniform wire of resistance 18Ω is stretched uniformly to triple its original length. The wire is then cut into three equal parts and connected in parallel across an ideal cell. The equivalent resistance of the parallel combination is:

- a) 2Ω . b) 6Ω c) 18Ω d) 54Ω

28. Ten identical cells, each of electromotive force E and internal resistance r , are connected in series to form a closed loop circuit loop. If two cells are found to be mistakenly connected with reversed polarities, the net effective EMF of the loop combination is:

- a) $10 E$. b) $8 E$ c) $6 E$ d) $4 E$

29. Two electric bulbs rated 220V, 40W and 220V, 60W are connected in series across a steady domestic 220V supply line. The total combined power consumed by the system is:

- a) 100 W. b) 50 W. c) 24 W d) 12 W

30. A galvanometer has a coil resistance of 50Ω and shows a full-scale deflection for a current of 2 mA. To convert this meter into a voltmeter capable of measuring up to 20 V, the required series resistance component is:

- a) 9950 Ω . b) 9900 Ω . c) 10050 Ω d) 4950 Ω

31. A straight horizontal wire of mass per unit length 50 g/m carries a current of 2 A. What minimum vertical magnetic field B must exist to completely suspend the conductor against gravity ($g = 10 \text{ m/s}^2$)?

- a) 0.25 T. b) 0.125 T. c) 0.50 T d) 1.00 T

32. Two long parallel copper wires separated by a distance of 10 cm carry currents of 4 A and 6 A in the opposite direction. The magnetic force per unit length acting between the conductors is:

- a) $4.8 \times 10^{-5} \text{ N/m}$, Attractive c) $2.4 \times 10^{-5} \text{ N/m}$, Attractive
b) $4.8 \times 10^{-5} \text{ N/m}$, Repulsive d) $2.4 \times 10^{-5} \text{ N/m}$, Repulsive

33. A aircraft with a wing span length of 40 m flies horizontally toward the north at a velocity of 1000 km/h. If the vertical component of the Earth's magnetic field at that site is $3.0 \times 10^5 \text{ T}$, the motional EMF induced across the wing tips is:

- a) 0.33 V. b) 0.44 V c) 1.20 V d) 0.11 V

34. An alternating current voltage supply is expressed mathematically by the function $V = 200\sqrt{2} \sin(100\pi t)$. The root-mean-square (rms) voltage and the frequency of the alternating source are:

- a) 200 V, 100 Hz c) 200 V, 50 Hz
b) $200\sqrt{2} \text{ V}$, 50 Hz d) 100 V, 50 Hz

35. Two point charges $+q$ and $-q$ are placed at a distance d apart. A third point charge Q is

placed on the perpendicular bisector of the line joining the two charges at a distance x from the center. If $x \gg d$, the electrostatic force acting on Q is proportional to:

- a) $1/x$ b) $1/x^2$ c) $1/x^3$ d) $1/x^4$

36. A solid non-conducting sphere of radius R is charged uniformly throughout its volume with a volume charge density ρ . The electric field intensity at a point situated at a radial distance r ($r < R$) from the center is:

- a) $\rho r / (3\epsilon_0)$. b) $\rho R^2 / (3\epsilon_0 r)$. c) $\rho R^3 / (3\epsilon_0 r^2)$ d) Zero

37. A $4 \mu\text{F}$ capacitor is charged to 50 V and then connected in parallel with an uncharged $6 \mu\text{F}$ capacitor. The common potential across the parallel combination stabilizes at:

- a) 20 V. b) 25 V c) 30 V d) 50 V

38. A parallel plate capacitor is charged by a battery and then disconnected from it. If the plates are now pulled further apart using insulated handles, what happens to the charge Q on the plates and the potential difference V across them?

- a) Q remains constant, V increases c) Q decreases, V remains constant
b) Q remains constant, V decreases d) Both Q and V increase

39. A photon of energy E collides elastically with a stationary free electron. The maximum possible kinetic energy transferred to the recoiling electron during Compton scattering is bounded when the photon scatter angle is:

- a) 0° b) 90° c) 180° (Backscattering) d) 45°

40. In the Bohr model of a hydrogen atom, the ratio of the period of revolution of an electron in the ground state ($n = 1$) to that in the first excited state ($n = 2$) is:

- a) 1 : 4 b) 1 : 2. c) 1 : 8 d) 1 : 16

41. If the accelerating voltage of an X-ray tube is increased from 10 kV to 20 kV, the minimum cutoff wavelength of the continuous spectrum shifts by a factor of:

- a) $\sqrt{2}$ b) 2. c) 0.5 d) 0.25

42. The binding energy per nucleon for a deuteron (${}_1\text{H}^2$) is 1.1 MeV and for an alpha particle (${}_2\text{He}^4$) is 7.0 MeV. If two deuterons fuse together to form a single helium nucleus, the net energy released in the reaction is:

- a) 23.6 MeV b) 25.8 MeV. c) 28.0 MeV d) 11.4 MeV

43. A radioactive sample has an initial activity of A_0 at $t = 0$. If its half-life is T , the time required for its remaining activity to drop to $A_0 / 16$ is:

- a) 3 T b) 4 T c) 5 T d) 8 T

44. When a p-n junction diode is operated under forward bias, the width of the depletion layer region and the height of the barrier potential will respectively:

- a) Increase, Decrease c) Decrease, Decrease
b) Decrease, Increase d) Increase, Increase

45. A full-wave rectifier circuit operates on an alternating sinusoidal input mains supply frequency of 50 Hz. The fundamental ripple frequency present in the rectified output DC signal is:

- a) 50 Hz b) 100 Hz. c) 200 Hz d) 25 Hz

46. The digital output Y of a two-input logic gate system is 0 only when both inputs A and B

are 1. For all other input configurations, the output Y is 1. This gate corresponds to a standard:

- a) AND Gate b) NOR Gate. c) NAND Gate d) XOR Gate

47. According to Moseley's law, the frequency (ν) of the characteristic K_α X-ray line emitted by a target element of atomic number Z is related by $\sqrt{\nu} = a(Z - b)$. The values of the screening constant b for K-series lines is:

- a) 1 b) 0. c) 7.4 d) 28

48. A radioactive mixture contains two unique nuclear species, A and B , with decay constants 10λ and λ respectively. Initially, they have the same number of nuclei. The time after which the ratio of the number of nuclei of A to that of B becomes $1/e^2$ is:

- a) $1/(9\lambda)$ b) $1/(11\lambda)$ c) $2/(9\lambda)$ d) $2/(11\lambda)$

49. A full-wave rectifier circuit operates on an alternating sinusoidal input mains supply frequency of 50 Hz. The fundamental ripple frequency present in the rectified output DC signal is:

- a) 50 Hz b) 100 Hz c) 200 Hz d) 25 Hz

50. A source of power P emits monochromatic light of wavelength λ . The total number of photons emitted per second by this source can be expressed as:

- a) $P\lambda / (hc)$ b) Phc / λ c) $hc / (P\lambda)$ d) $P\lambda hc$

51. If vapour density of the compound X_nO is 14, X has volume double that of Oxygen. What is the wt of X ?

- a) 6 b) 12 c) 8 d) 14

52. The number of moles of hydrogen molecules required to produce 20 moles of ammonia through Haber's process is:

- a) 40 b) 10 c) 20 d) 30

53. According to Bohr's theory, the angular momentum of an electron in the 5th orbit of a hydrogen atom is:

- a) $2.5 h/\pi$ c) $10 h/\pi$
b) $1.0 h/\pi$ d) $2.5 \pi/h$

54. The solubility of $BaSO_4$ in water is 2.42×10^{-3} g/L at 298 K. The value of its solubility product (K_{sp}) will be (Molar mass of $BaSO_4 = 233$ g/mol):

- a) 1.08×10^{-10} mol²L⁻² c) 1.08×10^{-12} mol²L⁻²
b) 1.08×10^{-14} mol²L⁻² d) 1.08×10^{-8} mol²L⁻²

55. Which of the following solutions will have the highest pH value?

- a) 0.01 M NaOH c) 0.01 M CH_3COOH
b) 0.01 M HCl d) 0.01 M NaCl

56. Find the pH of a buffer solution prepared by mixing 50 mL of 0.2 M CH_3COOH and 50 mL of 0.2 M CH_3COONa (pK_a of acetic acid = 4.74):

- a) 4.74 c) 3.74
b) 5.74 d) 7.00

57. For an ideal gas expanding into a vacuum under completely adiabatic conditions, which of the following sets of parameters is correct?

- a) $q = 0, \Delta T < 0, w > 0$ c) $q \neq 0, \Delta T = 0, w = 0$
b) $q = 0, \Delta T = 0, w = 0$ d) $q = 0, \Delta T = 0, w \neq 0$

58. If the rate constant for a first-order reaction is k , the time (t_{99.9%}) required for the completion of 99.9% of the reaction is given by:

- a) $t = 6.909 / k$ c) $t = 2.303 / k$
b) $t = 4.606 / k$ d) $t = 0.693 / k$

59. Radioactive decay follows which order of kinetics?

- a) Zero order c) Second order
b) First order d) Third order

60. In which of the following compounds does nitrogen exhibit its highest oxidation state?

- a) N_2H_4 c) N_3H
b) NH_3 d) NH_2OH

61. Two flasks A and B of equal volume are filled with SO_2 and He gases respectively at the same-day temperature and pressure. The ratio of the root mean square velocity (v_{rms}) of SO_2 to that of He is:

- a) 1:4 c) 1:16
b) 4:1 d) 1:2

62. Which of the following aqueous solutions should have the highest boiling point?

- a) 1.0 M NaOH c) 1.0 M NH_4NO_3
b) 1.0 M Na_2SO_4 d) 1.0 M KNO_3

63. During the electrolysis of molten Al_2O_3 , the number of Faradays required to deposit 9 g of Aluminum (Atomic mass = 27) at the cathode is:

- a) 1 c) 3
b) 2 d) 0.5

64. If the specific conductance of a 0.01 M weak monobasic acid solution is 2×10^{-4} S cm⁻¹ and its limiting molar conductivity is 400 S cm²mol⁻¹, the dissociation constant (K_a) of the acid is:

- a) 2.5×10^{-5} c) 5×10^{-4}
b) 2.5×10^{-6} d) 1.25×10^{-5}

65. For a zero-order reaction, the plot of concentration of reactant ($[R]$) versus time (t) yields a straight line with a slope equal to:

- a) $-k$ c) $-k / 2.303$
b) $+k$ d) $k / 2.303$

66. Which of the following properties is not shown by lyophilic sols?

- a) High hydration of particles
b) Reversible nature
c) Easy coagulation by adding small electrolytes
d) High viscosity compared to the medium

67. The vapor pressure of a liquid inside a closed container depends significantly on:

- a) The total surface area of the liquid exposed
b) The volume of the container
c) The temperature of the system and intermolecular
d) The amount of liquid left in the container

68. The correct order of atomic radii for the following isoelectronic species O^{2-} , F^- , Na^+ , Mg^{2+} is:

- a) $Mg^{2+} < Na^+ < F^- < O^{2-}$
b) $Mg^{2+} < F^- < Na^+ < O^{2-}$
c) $O^{2-} < F^- < Na^+ < Mg^{2+}$
d) $F^- < O^{2-} < Mg^{2+} < Na^+$

69. Which of the following molecules has a formal charge of +1 on the central oxygen atom?

- a) O_3 b) H_2O c) O_2 d) H_2O_2

70. In the extraction of copper from copper pyrites, the furnace configuration utilizes silica lines because it acts as:

- a) An acidic flux to remove FeO slag c) A reducing agent for Cu_2O
b) A basic flux to remove alumina d) An oxidizing agent for iron sulfides

71. The temporary hardness of water is caused due to the presence of:

- a) Calcium and magnesium bicarbonates c) Calcium and magnesium sulfates
b) Calcium and magnesium chlorides d) Sodium and potassium carbonates

72. Diborane (B_2H_6) features an interesting electron deficiency. It contains:

- a) Four 2-centre-2-electron bonds and two 3-centre-2-electron bonds
b) Two 2-centre-2-electron bonds and four 3-centre-2-electron bonds
c) Six 2-centre-2-electron bonds
d) Three 3-centre-2-electron bonds

73. The correct order of acid strength of the following nitrogen oxides is:

- a) $N_2O < NO < N_2O_3 < NO_2 < N_2O_5$ c) $N_2O_5 < NO_2 < N_2O_3 < NO < N_2O$
b) $NO < N_2O < N_2O_3 < NO_2 < N_2O_5$ d) $NO_2 < NO < N_2O < N_2O_3 < N_2O_5$

74. Bleaching powder is prepared by passing chlorine gas through:

- a) Slaked lime [$Ca(OH)_2$] c) Limestone [$CaCO_3$]
b) Quick lime [CaO] d) Calcium chloride solution

75. When acidified $K_2Cr_2O_7$ reacts with H_2S gas, the color changes to green because of the formation of:

- a) Cr^{3+} ions c) Chromic acid trace loops

b) Elemental sulfur residue

d) CrO₅ peroxide

76. Eutrophication of water bodies leading to depletion of dissolved oxygen content is caused by:

a) Excessive deposition of nutrient phosphate and nitrate

b) Acid rain runoff adding heavy metals fertilizers

c) Spillages of industrial oil pollutants

d) Accumulation of non-biodegradable plastics

77. Which of the following salt solutions will undergo both cationic and anionic hydrolysis, and its pH value will be independent of the concentration of the salt?

a) NaCl

c) CH₃COONH₄

b) NH₄Cl

d) CH₃COONa

78. In Lassaigne's test for the detection of nitrogen in an organic compound, the appearance of a Prussian blue color is due to the formation of:

a) Fe₄[Fe(CN)₆]₃

c) Fe₃[Fe(CN)₆]₂

b) Na₄[Fe(CN)₆]

d) Fe(CNS)₃

79. What is the total number of structural acyclic isomers possible for an organic compound having the molecular formula C₄H₁₀O?

a) 7

b) 4

c) 5

d) 6

80. Which of the following free radical species is the most stable due to maximum hyperconjugation and inductive effect?

a) (CH₃)₃C•

b) (CH₃)₂CH•

c) CH₃CH₂•

d) CH₃•

81. Which of the following organic compounds will exhibit optical isomerism (is chiral) due to the presence of an asymmetric carbon atom?

a) 2-Chlorobutane.

c) 1-Chlorobutane

b) 2-Chloropropane

d) 2-Methylpropane

82. Ozonolysis of an alkene followed by reduction with Zn/H₂O gives a mixture of methanal and ethanal. The alkene is:

a) Propene.

b) But-1-ene

c) But-2-ene

d) Pent-2-ene

83. Which of the following structures is not aromatic according to Hückel's Rule?

a) Cycloheptatrienyl cation (Tropylium ion).

c) Cyclooctatetraene

b) Cyclopentadienyl anion

d) Naphthalene

84. When chlorobenzene is treated with sodium metal in dry ether, diphenyl is formed. This synthetic reaction is known as:

a) Fittig reaction

c) Wurtz reaction

b) Wurtz-Fittig reaction

d) Frankland reaction

85. The low reactivity of aryl halides toward nucleophilic substitution reactions compared to alkyl halides is attributed to:

a) Partial double bond character of C-X bond via resonance

c) Stability of aryl carbocations

b) sp³ hybridization of the carbon bonded to halogen

d) Inductive electron withdrawal by the aromatic loop

86. Phenol is treated with chloroform and aqueous sodium hydroxide followed by acidification to give salicylaldehyde. This reaction is known as:

a) Reimer-Tiemann reaction.

c) Kolbe's reaction

b) Gattermann reaction.

d) Friedel-Crafts acylation

87. The reduction of a carbonyl group (>C=O) to a methylene group (-CH₂-) using zinc amalgam (Zn-Hg) and concentrated hydrochloric acid (HCl) is called:

a) Clemmensen reduction

c) Rosenmund reduction

b) Wolff-Kishner reduction

d) Stephen's reduction

88. Carboxylic acids react with halogen atoms in the presence of red phosphorus to introduce a halogen at the α-position. This reaction is known as:

a) Hell-Volhard-Zelinsky (HVZ) reaction.

c) Hunsdiecker reaction

b) Hoffmann bromamide degradation.

d) Schotten-Baumann reaction

89. The reaction of a primary amide with bromine and aqueous sodium hydroxide to yield a primary amine containing one less carbon atom is called:

a) Hoffmann bromamide degradation.

c) Carbylamine reaction

b) Gabriel phthalimide synthesis.

d) Curtius rearrangement

90. Which of the following disaccharides is a non-reducing sugar because its reducing groups are involved in glycosidic linkage?

a) Sucrose.

b) Lactose

c) Maltose

d) Cellobiose

91. Nylon-6,6 is a synthetic polyamide copolymer. Its constituent monomers are:

a) Hexamethylenediamine and Adipic acid.

b) Ethylene glycol and Phthalic acid.

c) Caprolactam molecules exclusively

d) Styrene and 1,3-Butadiene tracking tracks

92. Which of the following compound drugs acts as a broad-spectrum antibiotic capable of destroying multiple bacterial strains?

a) Chloramphenicol.

c) Penicillin G

b) Aspirin analgesics.

d) Prontosil azo structures

93. Hinsberg's reagent reacts with primary and secondary amines but leaves tertiary amines untouched. This reagent chemically corresponds to:

a) Benzenesulfonyl chloride

c) Acetyl chloride

b) p-Toluenesulfonate salt.

d) Thionyl chloride

94. Which of the following compounds reacts fastest via an S_N2 pathway with an aqueous nucleophile?

a) CH₃-I

b) CH₃-Br.

c) CH₃-Cl

d) CH₃-F

95. Which type of specialized glassware features a high silicon dioxide and boron trioxide content, imparting low coefficients of thermal expansion and high resistance to thermal shock?

a) Borosilicate glass (Pyrex)

c) Soda-lime soft glass

b) Lead crystal glass.

d) Safety laminated glass

96. The primary industrial catalyst configuration and promoter sequence used in Haber's process for the synthesis of ammonia gas corresponds to:

a) Finely divided iron (Fe) with molybdenum (Mo) as

b) Platinized asbestos beds with nitric oxide particles promoter

c) Vanadium pentoxide (V₂O₅) on silica lines

d) Raney nickel alloy working alongside copper dust

97. Which of the following chemical substances acts as an ideal primary standard in acid-base volumetric titrations due to its high purity, air stability, and non-hygroscopic nature?

a) Anhydrous Sodium Carbonate (Na₂CO₃)

c) Sodium Hydroxide pellets (NaOH)

b) Concentrated Hydrochloric Acid (HCl)

d) Hydrated Calcium Chloride flakes

98. Which spectroscopic tool relies upon transitions between nuclear spin states in a strong magnetic field to determine the carbon-hydrogen skeletal connectivity of organic molecules?

a) Nuclear Magnetic Resonance (NMR) Spectroscopy

b) Infrared (IR) Vibrational Spectroscopy

c) Ultraviolet-Visible (UV-Vis) Spectroscopy

d) Mass Spectrometry fragmentation

99. The closeness of a sequence of replicate analytical measurements to each other, irrespective of whether they hit the true theoretical target value, is defined as:

a) Precision.

c) Determinate bias error

b) Accuracy.

d) Absolute systemic drift

100. Sodium nitroprusside solution reacts with a solution containing sulfide radicals (S₂⁻) to yield a highly characteristic coloration. The color observed is:

a) Purple / Violet.

c) Blood Red

b) Canary Yellow

d) Prussian Blue

101. Beak in birth had teeth in

a) Ostrich.

b) Kiwi.

c) pelican

d) Archaeopteryx

102. Swabhav went to the British museum and found the skull having the cranial capacity of 900 cc. Identify the skull

a) Homo erectus pekinensis

c) Homo neanderthalis

b) Homo habilis.

d) Java man

103. Miller's experiment provided the evidence for theory of

a) Special creation.

c) Biogenesis

b) Abiogenesis.

d) organic evolution

104. Mr. Dangi went to dentist for dental caries and Doctor found to have dental formula of Dangi as 2123/2123. What is his estimated age?

a) 7 yrs.

b) 25 yrs.

c) 19 month

d) 15 yrs

105. Sohan went to tuth hospital with the complain of step ladder pattern fever , rose spot in chest ?on diagnosis he was having fever .which part of gut is involved

- a) Duodenum. b) Ileum. c)stomach d) illium

106. What is the thickness of respiratory membrane

- a) 20 mm. b) 0.5 micron. c)0.05micrometer d) 5 micron

107. Which of the following structure's paralysis need an immediate mechanical ventilation

- a) Diaphragm. c) Internal intercostal
b) External intercostal. d) Rectus abdominis

108. The first branch of human aorta is

- a) Brachiocephalic artery. c)Right subclavian
b) Coronary artery. d)left carotid

109. Niruta went to manmohan heart center for the cardiac examination .on examination doctor heard the abnormal heart sound . Such sounds are called

- a) pleural rub. b) Dub. c)lub d) Heart murmur

110. Tree of Life is called

- a) Arbor vitae. b) Organ of corti. c)pons varoli d) Diencephalon

111. Depolarization is due to

- a) efflux of Na⁺ ion c) Influx of Na⁺ ion
b) Efflux of K⁺ ion. d) Influx of K⁺ ion

112. The blood leaving the liver and joining towards the heart is rich in

- a) Bile. b) Urea. c)Glucose d)RBC

113. Longest loop of Henle is found in

- a) Kangaroo rat. c)opposum
b) Rhesus monkey. d)all of the above

114. The gestation period for human being is

- a) 300 days. b) 280 days. c)265 days d)240 days

115.Name the ion responsible for unmasking active sites on actin during muscle contraction

- a) Calcium b) Magnesium c) Sodium d) Potassium

116.Dipin, a medical student, notices that he is passing large amounts of very dilute urine throughout the day. He feels thirsty all the time and keeps drinking water, but his symptoms persist. On investigation, doctors find that his body is producing insufficient ADH (vasopressin).

This deficiency is most likely to result in:

- a) Diabetes mellitus disorder. c) Diabetes insipidus disorder
b) Goiter condition. d) Acromegaly presentation

117.Sertoli cells found in human testes provide nutritive support to:

- a) Leydig cells c) Developing spermatozoa
b) Spermatogonia cells d) Interstitial tissue

118.Photosensitive compound in human eye is made up of:

- a) Guanosine and Retinol. c) Opsin and Retinal
b) Opsin and Retinol. d) Transducin and Retinene

119.Which of the following options represents a pseudocoelomate animal condition?

- a) Annelida b) Arthropoda c) Aschelminthes d) Chordata

120. In case of poriferans, the spongocoel is lined with flagellated cells called:

- a) Oscula b) Porocytes c) Choanocytes d) Ostia

121.Lamprey (Petromyzon) and Hagfish (Myxine) differ from true fishes because they

- a) Lack fins. c) Lack scales and jaws
b) Possess a cartilaginous skeleton. d) Are exclusively freshwater form

122.In which of the following groups does the notochord persist throughout life and extend completely from the head to the tail region?

- a) Urochordata. c) Cephalochordata
b) Hemichordata. d) Vertebrata

123.Suppose a lab technician accidentally removes the chloragogen cells from an earthworm's coelomic cavity. The physiological crisis that follows is most functionally identical to losing which organ in a human?

- a) The kidney b) The lung c) The liver d) The pancreas

124. While mapping out the vascular pathway of an earthworm, a student notices that the direction of blood flow in a specific major vessel is strictly from the posterior end forward to the anterior end. Identify this vessel:

- a) Ventral vessel c) Lateral-esophageal vessel
b) Dorsal vessel d) Subneural vessel

125.The urogenital system of a male frog shows a close structural link between excretion and reproduction. Vasa efferentia enter the kidneys and open directly into:

- a) Bidder's canal b) Cloaca c) Urinary bladder d) Ureter

126.While observing a living frog, you notice it frequently lowers and raises the floor of its closed mouth cavity. What biological process is it executing?

- a) Regulating pressure in the middle ear
b) Moving air into and out of the lungs
c) Forcing food down the esophagus
d) Collecting chemical scents on its tongue

127.A patient is infected with Plasmodium falciparum, the most dangerous malarial species. This parasite causes a specific pattern of fever known clinically as:

- a) Benign tertian malaria c) Malignant tertian malaria
b) Quartan malaria d) Mild quotidian malaria

128. If a drug completely stops the development of Plasmodium gametocytes inside human blood vessels, what is the epidemiological impact of this intervention?

- a) The infected person is cured immediately
b) The patient will no longer suffer from fever or chills
c) The patient cannot pass the infection to a mosquito
d) The liver stage (schizogony) will accelerate

129.What type of connective tissue is primarily found forming the internal framework of lymphoid organs like the spleen and lymph nodes?

- a) Dense regular connective tissue c) Adipose tissue
b) Reticular connective tissue d) Areolar tissue

130.Mast cells are specialized cells of connective tissue proper that are found actively secreting:

- a) Hemoglobin, Myoglobin, and Melanin
b) Trypsin, Amylase, and Lipase
c) Histamine, Serotonin, and Heparin
d) Collagen, Elastin, and Reticulin

131. The urinary bladder must change its shape and stretch substantially when filled with urine without tearing. Which specialized epithelium enables this high level of distensibility?

- a) Keratinized stratified squamous epithelium
b) Simple columnar epithelium
c) Transitional epithelium
d) Non-ciliated pseudostratified epithelium

132.While analyzing a tissue section under a microscope, a student notices concentric ring configurations called lamellae surrounding a central canal containing blood vessels. Identify the cell type residing within the lacunae of this tissue:

- a) Chondrocytes c) Osteocytes
b) Fibrocytes d) Adipocytes

133. Vaccines induce a safe simulation of a microbial attack. If a child receives an injection of a purified bacterial toxoid (inactivated exotoxin), what is the exact nature of the immunity generated?

- a) Natural active immunity c) Artificial active immunity
b) Natural passive immunity d) Artificial passive immunity

134. Human Immunodeficiency Virus (HIV) selectively targets and destroys which specific cellular component of the human immune system, leading to AIDS?

- a) B-lymphocytes
- b) Cytotoxic T-cells (T-C)
- c) Helper T-cells (TH)
- d) Natural Killer cells (NK cells)

135. The diagnostic confirmation of Typhoid fever in a clinical laboratory is reliably achieved through which of the following tests?

- a) Schick test
- b) Dick test
- c) Widal test
- d) Mantoux test

136. Which of the following anti tubercular drug causes ottotoxicity ?

- a) Streptomycin.
- b) Rifampicin
- c) Ethambutol
- d) Isoniazide

137. The "Golden Rice" variant was successfully engineered by medical botanists to serve as a bio-fortified crop that directly addresses clinical deficiencies of which essential nutrient?

- a) Vitamin C
- b) Iron
- c) Vitamin A (Beta-carotene)
- d) Vitamin B12

138. While analyzing an X-ray radiograph of a fractured femur, Birendra notes that dense bone structures appear white, while surrounding muscle tissues look dark grey. What physical process explains this difference?

- a) Muscles absorb X-rays via dense magnetic fields
- b) Calcium in bone has a high atomic number and attenuates X-rays
- c) Bone tissue selectively reflects X-rays like a mirror
- d) Soft tissues accelerate the wavelength of incoming X-ray photons

139. The pyramid of biomass in a deep aquatic ecosystem (such as a large lake or ocean) is typically:

- a) Always upright
- b) Bell-shaped
- c) Inverted
- d) Spindle-shaped

140. According to the Red Data Book maintained by the IUCN, a species facing an extremely high risk of extinction in the wild in the immediate future is categorized as:

- a) Vulnerable
- b) Critically Endangered
- c) Endangered
- d) Extinct in the Wild

141. Which of the following statements is strictly TRUE regarding the structure of a plant cell wall?

- a) The primary wall is static and never diminishes
- b) The middle lamella is chiefly made of calcium pectate
- c) Secondary wall is formed on the outer side of primary wall
- d) Cellulose is entirely absent in the primary cell wall

142. Which of the following cell organelles is responsible for synthesizing and storing non-pigmented nutrients like starch in potato tubers?

- a) Elaioplasts
- b) Aleuoplasts
- c) Amyloplasts
- d) Chromoplasts

143. Consider a plant species where fruit weight is controlled by three polygenic gene pairs (A, B, C). Homozygous aabbcc yields a 10g fruit, while AABBCC yields a 40g fruit. What is the weight contribution of each dominant allele?

- a) 10g
- b) 30g
- c) 5g
- d) 15g

144. If a template strand of plant DNA displays the linear sequence 5'-ATCGGTA-3', what will be the exact sequence of the complementary mRNA strand synthesized during transcription?

- a) 5'-UAGCCUA-3'
- b) 3'-UAGCCAU-5'
- c) 5'-UACCOAU-3'
- d) 5'-TACCGAT-3'

145. Which of the following nitrogenous bases is strictly unique to the structure of RNA molecules, serving as a replacement for Thymine?

- a) Adenine
- b) Cytosine
- c) Uracil
- d) Guanine

146. The experimental phenomenon where an F1 individual shows a phenotype that perfectly blends or sits intermediate between both homozygous parents is known as:

- a) Codominance
- c) Incomplete dominance

- b) Epistasis
- d) Pleiotropy

147. The chromosomal mechanism of inheritance was experimentally verified and structurally linked using Drosophila and sweet pea models by:

- a) Gregor Johann Mendel
- b) Watson and Crick
- c) Thomas Hunt Morgan
- d) Hugo de Vries

148. Which specific non-sense or stop codon halts the translation mechanism of polypeptides across almost all plant cellular ribosomes?

- a) AUG
- b) GUG
- c) UAA
- d) UGG

149. Stomata which are structurally distributed equally on both the upper adaxial and lower abaxial epidermis of a leaf are typically seen in:

- a) Dorsiventral dicotyledonous leaves
- b) Submerged hydrophytic leaves
- c) Isobilateral monocotyledonous leaves
- d) Xerophytic pine needles

150. The cork cambium (phellogen) undergoes periclinal divisions to produce phellem on its outer side and phelloderm on its inner side. Phellem cells are waterproof due to the deposition of:

- a) Lignin
- b) Cutin
- c) Suberin
- d) Hemicellulose

151. In a primary dicot root, the protoxylem elements are located closer to the periphery (pericycle), while the metaxylem elements face the center. This pattern of development is termed:

- a) Endarch
- b) Mesarch
- c) Exarch
- d) Centrifugal

152. The terminal electron acceptor of the mitochondrial electron transport chain (ETC) operating during aerobic respiration is:

- a) Cytochrome c
- b) NAD⁺
- c) Molecular Oxygen (O₂)
- d) Ubiquinone

153. The "Richmond-Lang effect," which denotes the striking delay of leaf senescence via nutrient mobilization, is a classic feature of:

- a) Auxins
- b) Gibberellins
- c) Cytokinins
- d) Absciscic acid

154. A mutant strain of barley seeds fails to produce alpha-amylase when treated with standard concentrations of Gibberellins. Which specific cellular layer of the seed is likely defective or missing?

- a) Endosperm matrix
- b) Coleoptile tip
- c) Aleurone layer
- d) Scutellum

155. The conversion of one molecule of fructose 1,6-bisphosphate into two molecules of pyruvate during glycolysis yields a NET of how many ATP?

- a) 2 ATP
- b) 36 ATP
- c) 4 ATP
- d) 38 ATP

156. Which tool is standardly employed to measure the rate of transpiration in a lab setting by monitoring the movement of an air bubble?

- a) Auxanometer
- b) Respirometer
- c) Ganong's Potometer
- d) Porometer

157. If a plant geneticist isolates a mutant rice strain that exhibits "foolish seedling" disease (extreme over-elongation) even in the complete absence of the fungal pathogen Gibberella fujikuroi, the mutation most likely:

- a) Inactivates the plant's native auxin receptors
- b) Causes constitutive (permanent) activation of the gibberellin signaling pathway
- c) Blocks the synthesis of absciscic acid
- d) Deletes the gene for alpha-amylase production

158. Which of the following cellular organelle pairs is entirely bound by a single membrane envelope system?

- a) Mitochondria and Lysosomes
- b) Chloroplasts and Peroxisomes
- c) Lysosomes and Peroxisomes
- d) Nuclei and Ribosomes

159. During the cell cycle, a critical checkpoint occurs at the end of G1 phase. If a cell fails to clear this point due to DNA damage, it enters:

- a) S phase directly
- c) Quiescent stage (G₀ phase)

b) G₂ phase prematurely d) Mitotic prophase symmetry
160. A specialized epithelial cell needs to transport glucose against its concentration gradient into the cytoplasm. It uses an influx of sodium ions moving down their electrochemical gradient to power this. This mechanism is an example of:

- a) Primary active transport c) Secondary active symport
 b) Simple facilitated diffusion d) Receptor-mediated pinocytosis

161. If a cell has a diploid chromosome number of 2n=16, how many total DNA molecules (chromatids) are present in the cell during the G₂ phase of the cell cycle?

- a) 8 c) 32
 b) 16 d) 64

162. Ribosomal RNA (rRNA) is actively synthesized and assembled into ribosomal subunits inside which specific nuclear sub-domain?

- a) Kinetochore c) Nucleolus
 b) Nuclear pore complex d) Perinuclear space matrix

163. If a cell biologist stains a green algal cell and uncovers a proteinaceous core surrounded by a starch sheath inside the chloroplast, they have mapped a:

- a) Carotenoid crystal c) Pyrenoid
 b) Gas vacuole d) Eyespot matrix

164. You are tracking a moss spore after it lands on damp soil. Before developing into a mature leaf-bearing gametophore, it forms a creeping, green, branched filamentous stage called a: [C

- a) Prothallus c) Protonema
 b) Sorus d) Primary endosperm

165. In Equisetum (horsetail), the sporangia are organized on compact, distinct terminal structures known as:

- a) Sori c) Strobili or Cones
 b) Indusia d) Sporophyll bands

166. A student looks at a pollen grain from Pinus under a microscope and notes two hollow, air-filled bladder extensions. These wings are an adaptation for:

- a) Insect-mediated pollination c) Anemophily (Wind pollination)
 b) Hydrophily (Water pollination) d) Epizoochory

167. During megasporogenesis in a typical angiosperm, a single functional megaspore develops into a mature embryo sac. This final structure is characterized as:

- a) 8-nucleate, 8-celled c) 7-celled, 8-nucleate
 b) 3-celled egg apparatus d) 4-nucleate macropore

168. A plant's life cycle features an alternation of generations where both the sporophyte and gametophyte are free-living, independent, and photosynthetic. This fits:

- a) A typical bryophyte c) A typical pteridophyte
 b) A naked gymnosperm tree d) A monocotyledonous herb

169. Tetradyneous stamens, featuring two short outer filaments and four long inner filaments (2 + 4), are a key diagnostic character of:

- a) Mustard family (Brassicaceae) c) Pea family (Fabaceae)
 b) Potato family (Solanaceae) d) Lily family (Liliaceae)

170. Mad Cow Disease (Bovine Spongiform Encephalopathy) and Kuru in humans are caused by slow-acting, infectious agents composed entirely of abnormal proteins called:

- a) Viroids c) Prions
 b) Virions d) Bacteriophages

171. When a T-even bacteriophage infects an E. coli cell, which specific part of the viral structure physically penetrates through the bacterial cell wall?

- a) The entire protein capsid along with the genome
 b) The tail fibers and baseplate assembly
 c) Only the viral nucleic acid genome
 d) Empty structural capsomeres

172. According to Alexander von Humboldt's classic ecological observations, species richness increases with increasing explored area, but only up to a limit.

This relationship plots as a straight line on a log-log scale described by which mathematical formula?

- a) $\log S = \log C + Z \log A$ c) $\log A = \log C + Z \log S$
 b) $\log S = \log Z + C \log A$ d) $\log S = \log C * Z \log A$

173. The association between sea anemones and hermit crabs, where the crab gains protection and the anemone achieves mobility without structural metabolic dependence, is:

- a) Amensalism c) Proto-cooperation (Facultative Mutualism)
 b) Parasitism d) Commensalism

174. In a pioneer ecological succession on bare rock (Lithosere), which organisms act as the crucial initial colonizers preparing the substrate for higher plants?

- a) Mosses c) Lichens
 b) Pteridophytes d) Green algae

175. According to Lindeman's foundational Efficiency Law of trophic dynamics, if primary producers trap 10,000J of net energy from sunlight, how much energy enters the secondary consumers?

- a) 1,000J c) 100J
 b) 10J d) 1J

176. In micropropagation via plant tissue culture, growing uniform virus-free clonal plants from a completely virus-infected parent plant can be successfully accomplished by using which explant?

- a) Leaf disc segments c) Apical and axillary meristems
 b) Vascular cambium tissue d) Mature cotyledons

177. Atropine, a drug used to treat bradycardia (slow heart rate) and certain poisoning cases, is extracted from Atropa belladonna. This plant belongs to which family?

- a) Fabaceae c) Solanaceae
 b) Brassicaceae d) Liliaceae

178. To bypass the natural species barrier in cross-hybridization, cell walls of two different plant species are digested using pectinase and cellulase. The resulting naked protoplasts are then fused using:

- a) Sodium carbonate c) Polyethylene glycol (PEG) or Electrofusion
 b) Ethidium bromide d) Colchicine chemical bath

179. Restriction enzymes are called:

- a) Molecular scissors. c) Molecular glue
 b) Molecular carriers. d) Molecular markers

180. Ti plasmid is obtained from:

- a) E. coli. c) Bacillus thuringiensis
 b) Agrobacterium tumefaciens. d) Rhizobium

181. If in a certain code language, the word 'SUBBA' is written as 'VXDED', how will the word 'KHARIDAR' be coded in that same system?

- a) NKDULGDU c) NDKULGDU
 b) NKDULGUD. d) MHKVLGDU

182. Arrange the following words in a meaningful logical sequence from microscopic to macroscopic scale:

1. District 2. Ward 3. Country 4. Province 5. Municipality

- a) 2, 5, 1, 4, 3. c) 2, 1, 5, 4, 3
 b) 5, 2, 1, 4, 3. d) 2, 5, 4, 1, 3

183. If 'Pen' is called 'Paper', 'Paper' is called 'Ink', 'Ink' is called 'Desk', and 'Desk' is called 'Office', where does a student fill a fountain pen from?

- a) Desk. c) Paper
 b) Ink d) Office

184. Pointing to a photograph, a woman says: "This man's son's sister is my mother-in-law." How is the woman's husband related to the man in the photograph?

- a) Great-grandson. c) Son
 b) Grandson. d) Nephew

185. Select the odd alphanumeric group out from the following structural choices:

- a) D4W23. c) H8S19
 b) F6U21 d) J10P16

186. Complete the missing entry in the following number series:

3, 5, 9, 17, 33, _____

- a) 65. c) 60
b) 52. d) 69

187. A man sells an article at a profit of 20%. If he had bought it for 20% less and sold it for Rs. 18 less, he would have gained 25%. Find the initial Cost Price (CP) of the article.

- a) Rs. 90 c) Rs. 85
b) Rs. 100 d) Rs. 120

188. The average age of a group of 8 officers increases by 2 years when two old officers aged 40

years and 42 years are replaced by two new trainee sub-officers. What is the average age of the two new trainees?

- a) 49 years. c) 45 years
b) 41 years. d) 51 years

189. A tank can be filled by Pipe A in 6 hours and emptied by Pipe B in 8 hours. If both pipes are opened simultaneously when the tank is empty, how long will it take to fill the tank completely?

- a) 24 hours. c) 14 hours
b) 12 hours. d) 48 hours

190. Ram can complete a piece of public office work in 10 days, while Shyam takes 15 days to do the same work. They start working together, but Ram leaves 2 days before the completion of the work. What is the total number of days taken to complete the entire assignment?

- a) 7.2 days. c) 6.5 days
b) 8.0 days. d) 7.0 days

191. A person starts walking from his office towards the East. After walking 10 meters, he turns right and walks 5 meters. Then he turns left and walks 10 meters. In which direction is he now facing relative to his starting point?

- a) South-East. c) East
b) North-East d) South

192. In a row of 35 successful candidates, Mahesh is ranked 14th from the top. What is his position rank when counted from the bottom of the row?

- a) 22nd. c) 21st
b) 23rd d) 20th

193. Choose the correct alternative that logical sequencing dictates to replace the missing segment: Z1A, X2D, V6G, T24J, _____

- a) R120M. c) R120N
b) S120M d) P120L

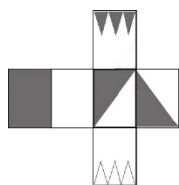
194. A scooter trekker leaves Kathmandu heading North towards Gosaikunda. He drives 30 km, turns left, and drives 15 km. He then turns left again and drives 30 km. How far and in which direction is he now located relative to his starting point?

- a) 15 km West. c) 15 km East
b) 30 km North. d) 0 km (Back at start)

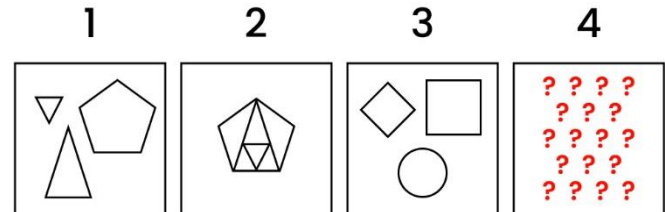
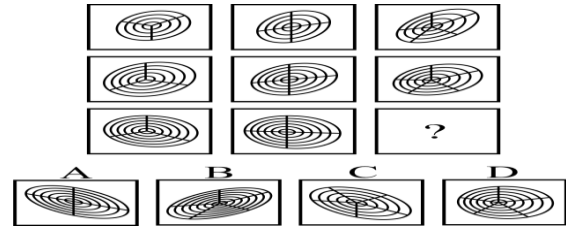
195. A clock is placed on a flat table. At 12:00 PM, its minute hand points North. At 3:45 PM, in which general compass direction will the minute hand be pointing?

- a) West c) East
b) South. d) North-West

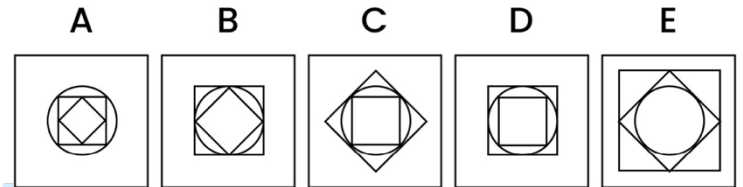
196. Which of the 4 possible options represent the cube in folded form



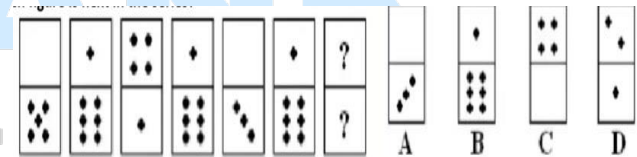
197. Select the suitable option that would complete the figure



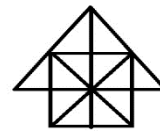
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199. Which figure is next in series



200. Count the number of triangle?



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– Best of Luck –

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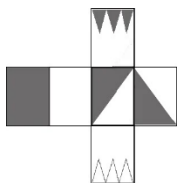
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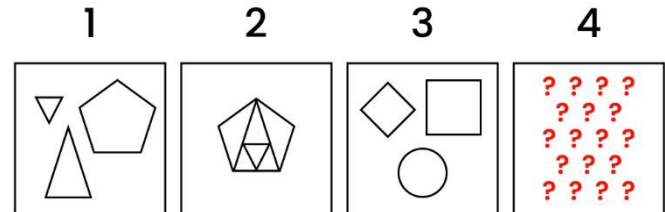
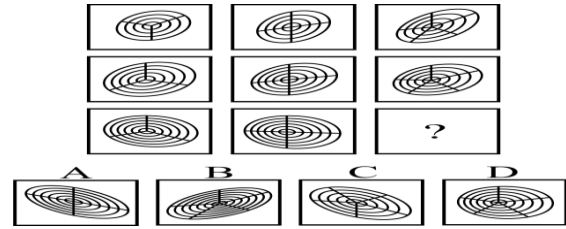
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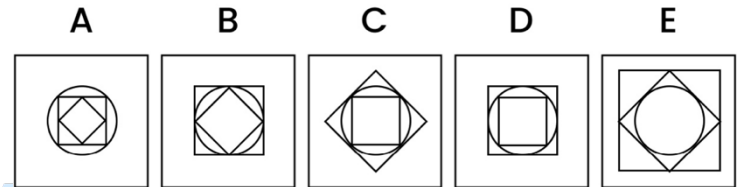
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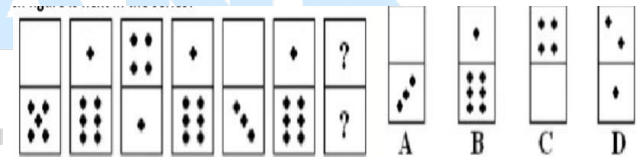
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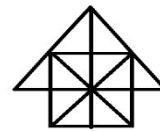
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